NewsBot Intelligence System — Reflective Journal

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Project Format: Individual project

Learning and Technical Experience

In this project, I worked on a NewsBot Intelligence System using the BBC News dataset. The goal was to combine different NLP techniques to classify news, analyze sentiment, identify grammatical patterns, and extract named entities. Working through the notebook helped me to understand how these techniques connect within one system.

I started with 1,490 articles across five categories: business, entertainment, politics, sport, and technology. During exploration, I found 50 duplicate texts. After removing them, 1,440 articles remained. They we are no missing values or empty articles. This showed me why checking data quality is important before training a model.

The preprocessing section helped me understand how cleaning changes text. I removed URLs, HTML, special characters, and stopwords, and applied basic lemmatization. The average article length decreased from about 385 to 210 words. I also noticed that pre-processing, can introduce mistakes, such as changing “boss” to “bos.” This taught me to inspect examples instead of assuming every transformation improves the data.

Integration Challenges and Results

One of the main challenges was adapting the template to the actual dataset. Some cells expected separate titles, date, and source columns, but BBC only provided article ID, text, and category. I adjusted those cells while keeping the original text available for sentiment, entity recognition, and grammatical analysis.

Another challenge was combining text features with sentiment scores and article length. I used a classification pipeline, so TF-IDF and numerical scaling were fitted using only training data. This also allowed preprocessing to happen separately within each cross-validation fold, helping prevent information from the test set from influencing training.

I compare Naive Bayes, Logistic Regression, and SVM. SVM had the highest training cross-validation accuracy and achieved 95.83% accuracy on the 288 test articles, correctly classifying 276 of them. Sports had a recall of 1.00, while technology had the lowest recall at approximately 0.89. These results showed that overall accuracy is useful, but category-level evaluation provides more detail about model weaknesses.

POS tagging, dependency parsing, and entity recognition also revealed limitations. Some grammatical labels and entity boundaries were incorrect. The dataset’s lowercase text made proper-noun recognition more difficult. These examples helped me to understand that successful execution does not guarantee correct language analysis.

Individual Organization and Contributions

I completed the project individually with my professor's approval. I was responsible for preparing the data, adapting the notebook, running the analyses, comparing models, and reviewing the integrated system. I organized the work following the notebook sections in order and checked outputs before moving forward.

I used AI assistance to understand code, troubleshoot errors, and adapt the template. Problems included missing files, indentation errors, and variables becoming unavailable after Colab restarted. These experiences taught me to keep the notebook organized, understand dependencies between cells, and rerun earlier sections when necessary. Working individually also made time management important because I handled all parts of the project.

Business Value and Ethical Considerations

NewsBot could support article organization, content routing, and searches for mentions of people, companies, and locations. Its sentiment analysis could help users explore patterns in article tone. However, the results could support human decisions rather than be treated as verified facts.

Testing synthetic examples demonstrated this limitation. The sports example was classified as sport, while the Microsoft example was ambiguous between technology and business. The climate example exposed weaknesses because the system had no environmental category, VADER assigned a positive sentiment score, and entity recognition produced an incorrect phrase. The SVM decision also differed from the category with the highest estimated probability.

These outcomes showed why model probabilities should not be interpreted as guarantees. Article sentiment also does not necessarily describe the author's opinion or sentiment toward a particular entity. Using these outputs without reviewing their limitations could mislead users.

Future Improvements and Professional Development

In future improvements I would include testing longer articles from different sources, comparing feature combinations, tuning model parameters, and improving lemmatization and entity recognition. I would also like to explore specific emotion analysis and transformer models that may handle context more effectively. These improvements would need evaluation before claiming better performance. This project helped me practice debugging, interpreting results, and connecting NLP techniques within an integrated prototype. For my portfolio, I want to show both the successful classification results and the errors found during testing.